Integration: Inverse Trigonometric Functions

Questions 1–51

Question 1. Use the substitution $x = \sin u$ to show that

$$\int \frac{1}{\sqrt{1-x^2}}\,dx = \arcsin x + c.$$

Question 2. Use the substitution $x = \cos u$ to show that

$$\int \frac{1}{\sqrt{1-x^2}}\,dx = -\arccos x + c.$$

Question 3. Find the exact value of

$$\int_{-1/2}^{1/2} \frac{1}{\sqrt{1-x^2}}\, dx.$$

Question 4. Use the substitution $x = \tan u$ to show that

$$\int \frac{1}{1+x^2}\, dx = \arctan x + c.$$

Question 5. Find the exact value of

$$\int_{-1}^{1} \frac{1}{1+x^2}\,dx.$$

Question 6. Find:

a) $\displaystyle\int \frac{2}{\sqrt{1-x^2}}\,dx$

b) $\displaystyle\int \frac{5}{1+x^2}\,dx$

c) $\displaystyle\int \frac{1}{\sqrt{1-9x^2}}\,dx$

d) $\displaystyle\int \frac{1}{1+4x^2}\,dx$

e) $\displaystyle\int \frac{1}{\sqrt{1-2x^2}}\,dx$

f) $\displaystyle\int \frac{1}{1+8x^2}\,dx$

g) $\displaystyle\int \frac{10}{\sqrt{1-5x^2}}\,dx$

h) $\displaystyle\int \frac{200}{1+10x^2}\,dx$

i) $\displaystyle\int \frac{4}{\sqrt{1-\frac{1}{4}x^2}}\,dx$

j) $\displaystyle\int \frac{18}{1+\frac{2}{3}x^2}\,dx$

Question 7. Find the exact value of:

a) $\displaystyle\int_0^{\sqrt{2}/8} \frac{1}{\sqrt{1-16x^2}}\,dx$

b) $\displaystyle\int_{1/(2\sqrt{3})}^{1/2} \frac{30}{\sqrt{1-3x^2}}\,dx$

Question 8. Find the exact value of:

a) $\displaystyle\int_0^{\sqrt{3}/6} \frac{1}{1+36x^2}\,dx$

b) $\displaystyle\int_{1/\sqrt{3}}^{1} \frac{12}{1+3x^2}\,dx$

Question 9.

a) Show that

$$\frac{1}{2+x^2} \equiv \frac{1}{2(1+(kx)^2)},$$

where k is a real number to be found.

b) Hence find $\int \frac{1}{2+x^2}\,dx$.

Question 10.

a) Show that

$$\frac{1}{\sqrt{2-x^2}} \equiv \frac{1}{\sqrt{2}\sqrt{1-(kx)^2}},$$

where k is a real number to be found.

b) Hence find $\int \frac{1}{\sqrt{2-x^2}}\,dx$.

Question 11. Find:

a) $\displaystyle\int \frac{1}{4+x^2}\,dx$

b) $\displaystyle\int \frac{1}{\sqrt{9-x^2}}\,dx$

c) $\displaystyle\int \frac{1}{5+x^2}\,dx$

d) $\displaystyle\int \frac{1}{\sqrt{8-x^2}}\,dx$

e) $\displaystyle\int \frac{4}{12+x^2}\,dx$

f) $\displaystyle\int \frac{10}{\sqrt{5-x^2}}\,dx$

g) $\displaystyle\int \frac{1}{2+4x^2}\,dx$

h) $\displaystyle\int \frac{1}{\sqrt{16-9x^2}}\,dx$

i) $\displaystyle\int \frac{2}{4+25x^2}\,dx$

j) $\displaystyle\int \frac{3}{\sqrt{12-\frac{1}{3}x^2}}\,dx$

Question 12. Find the exact values of:

a) $\displaystyle\int_{-\sqrt{3}/3}^{\sqrt{3}/3} \frac{1}{\sqrt{4-9x^2}}\,dx$

b) $\displaystyle\int_{0}^{\frac{5}{8}(\sqrt{5}-1)} \frac{3}{\sqrt{100-16x^2}}\,dx$

Question 13. Find the exact values of:

a) $\displaystyle\int_{-5\sqrt{3}/27}^{5\sqrt{3}/27} \frac{1}{25+81x^2}\,dx$

b) $\displaystyle\int_{0}^{3(\sqrt{2}-\sqrt{3/2})} \frac{9}{9+2x^2}\,dx$

Question 14. Given that a and b are constants, find:

a) $\displaystyle\int \frac{1}{\sqrt{a^2-x^2}}\,dx$

b) $\displaystyle\int \frac{1}{a^2+x^2}\,dx$

c) $\displaystyle\int \frac{1}{\sqrt{a^2-b^2x^2}}\,dx$

d) $\displaystyle\int \frac{1}{a^2+b^2x^2}\,dx$

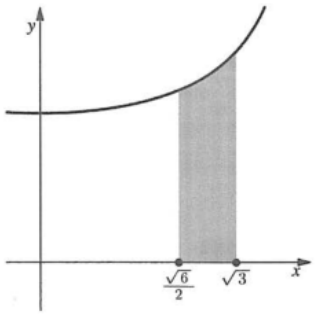
Question 15. Given that a and b are positive constants, find the exact value of

$$\int_0^{\sqrt{ab}} \frac{1}{\sqrt{b - \frac{1}{a}x^2}}\, dx.$$

Question 16. The curve shown in the diagram below has equation

$$y = \frac{6}{\sqrt{18 - 3x^2}}.$$

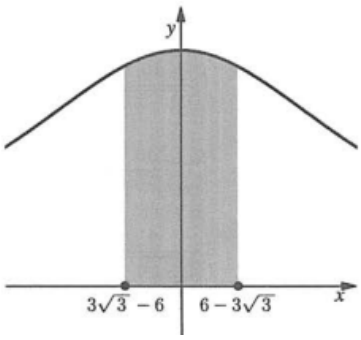
Find the exact area of the shaded region.



Question 17. The curve shown in the diagram below has equation

$$y = \frac{9}{36 + 4x^2}.$$

Find the exact area of the shaded region.



Question 18. It is given that

$$f(x) = \frac{19x^2 + 34}{(1 + x^2)(9 + 4x^2)}, \qquad x \in \mathbb{R}.$$

a) Write $f(x)$ in the form

$$\frac{Ax + B}{1 + x^2} + \frac{Cx + D}{9 + 4x^2},$$

where A, B, C and D are constants to be found.

b) Hence find $\int f(x)\,dx$.

Question 19. Find:

a) $\displaystyle\int \frac{107x^2 - 3}{(4 + 9x^2)(1 + 25x^2)}\, dx$

b) $\displaystyle\int \frac{8x^4 + 12x^3 + 33x^2 + 3x + 4}{(4 + x^2)(1 + 4x^2)}\, dx$

Question 20. Show that

$$\int_{1/2}^{\sqrt{3}/2} \frac{16x^2 + 5x + 4}{x(8x^2 + 2)}\, dx = a\pi + \ln b,$$

where a is a rational number and b is a positive integer to be found.

Question 21. Given that

$$\frac{1+x}{1+x^2} \equiv \frac{1}{1+x^2} + \frac{x}{1+x^2},$$

find $\displaystyle\int \frac{1+x}{1+x^2}\,dx$.

Question 22. Find:

a) $\displaystyle\int \frac{2x+1}{1+4x^2}\,dx$

b) $\displaystyle\int \frac{3+5x}{4+x^2}\,dx$

c) $\displaystyle\int \frac{2-3x}{9+4x^2}\,dx$

d) $\displaystyle\int \frac{5-4x}{25+16x^2}\,dx$

Question 23. Given

$$\frac{1+x}{\sqrt{1-x^2}} \equiv \frac{1}{\sqrt{1-x^2}} + \frac{x}{\sqrt{1-x^2}},$$

find $\displaystyle\int \frac{1+x}{\sqrt{1-x^2}}\,dx$.

Question 24. Find:

a) $\displaystyle\int \frac{3x-1}{\sqrt{1-9x^2}}\,dx$

b) $\displaystyle\int \frac{1+8x}{\sqrt{4-x^2}}\,dx$

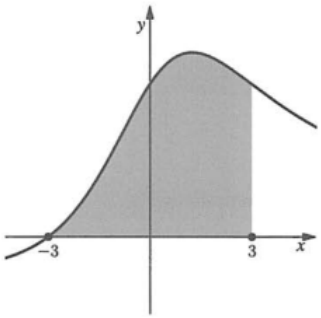
c) $\displaystyle\int \frac{6-x}{\sqrt{9-4x^2}}\,dx$

d) $\displaystyle\int \frac{10-9x}{\sqrt{4-25x^2}}\,dx$

Question 25. The curve shown in the diagram below has equation

$$y = \frac{3+x}{9+x^2}.$$

Find the exact area of the shaded region.



Question 26.

a) Find the general solution to the differential equation

$$\frac{dy}{dx} = \frac{x+1}{\sqrt{9-x^2}}.$$

b) Hence, find a particular solution such that $y = 3$ when $x = 0$.

Question 27.

a) Find $\displaystyle\int \frac{x}{\sqrt{25-x^2}}\,dx$.

b) Find $\displaystyle\int \frac{1}{\sqrt{25-x^2}}\,dx$.

c) It is given that

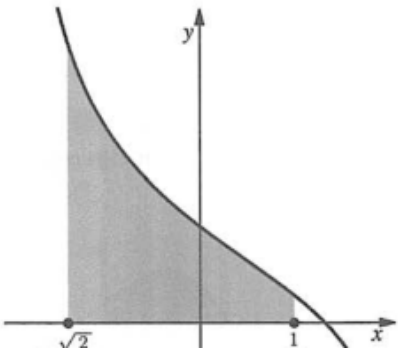
$$f(x) = \frac{5x+20}{\sqrt{25-x^2}}, \qquad |x| < 5.$$

Find the exact mean value of $f(x)$ over the interval $\left[-\frac{5}{2}, 0\right]$.

Question 28. The curve shown in the diagram below has equation

$$y = \frac{4-9x}{\sqrt{4-9x^2}}.$$

Find the exact area of the shaded region.



Question 29. For each of the following integrals, suggest a suitable trigonometric substitution and then integrate:

a) $\displaystyle\int \frac{1}{\sqrt{1-x^2}}\,dx$

b) $\displaystyle\int \frac{1}{1+x^2}\,dx$

c) $\displaystyle\int \frac{1}{\sqrt{1-4x^2}}\,dx$

d) $\displaystyle\int \frac{1}{1+9x^2}\,dx$

e) $\displaystyle\int \frac{1}{\sqrt{9-x^2}}\,dx$

f) $\displaystyle\int \frac{1}{4+x^2}\,dx$

g) $\displaystyle\int \frac{1}{\sqrt{1-25x^2}}\,dx$

h) $\displaystyle\int \frac{1}{1+81x^2}\,dx$

i) $\displaystyle\int \frac{1}{\sqrt{4-9x^2}}\,dx$

j) $\displaystyle\int \frac{1}{9+4x^2}\,dx$

k) $\displaystyle\int \frac{1}{\sqrt{25-4x^2}}\,dx$

l) $\displaystyle\int \frac{1}{49+36x^2}\,dx$

m) $\displaystyle\int \frac{1}{\sqrt{16-3x^2}}\,dx$

n) $\displaystyle\int \frac{1}{4+5x^2}\,dx$

o) $\displaystyle\int \frac{1}{\sqrt{9-81x^2}}\,dx$

Question 30. For each of the following integrals, suggest a suitable trigonometric substitution and then integrate:

a) $\displaystyle\int \sqrt{1-x^2}\,dx$

b) $\displaystyle\int (1-x^2)^{3/2}\,dx$

Question 31. For each of the following integrals, suggest a suitable trigonometric substitution and then integrate:

a) $\displaystyle\int \frac{1}{(1+x^2)^2}\,dx$

b) $\displaystyle\int \frac{1}{(4+x^2)^2}\,dx$

c) $\displaystyle\int \frac{1}{(1+x^2)^3}\,dx$

d) $\displaystyle\int \frac{1}{(9+4x^2)^3}\,dx$

Question 32. For each of the following integrals, suggest a suitable trigonometric substitution and then integrate:

a) $\displaystyle\int \sqrt{36-x^2}\,dx$

b) $\displaystyle\int (1-4x^2)^{3/2}\,dx$

Question 33. Complete the square for each of these:

a) $26 - 10x + x^2$

b) $101 + 20x + x^2$

c) $2 + 4x + 4x^2$

d) $5 - 12x + 9x^2$

e) $25 - 8x + x^2$

f) $29 + 8x + 4x^2$

Question 34.

a) Write $2 + 2x + x^2$ in the form $p + (x + q)^2$, where p and q are integers.

b) Hence, using the substitution $x + 1 = \tan u$, find $\displaystyle\int \frac{1}{2 + 2x + x^2}\, dx$.

Question 35. Find:

a) $\displaystyle\int \frac{1}{10-6x+x^2}\,dx$

b) $\displaystyle\int \frac{1}{65+16x+x^2}\,dx$

Question 36.

a) Write $10+18x+9x^2$ in the form $p+q(x+r)^2$, where p, q and r are integers.

b) Hence, using the substitution $3x+3=\tan u$, find $\displaystyle\int \frac{1}{10+18x+9x^2}\,dx$.

Question 37. Find:

a) $\displaystyle\int \frac{1}{10+12x+4x^2}\,dx$

b) $\displaystyle\int \frac{1}{50-28x+4x^2}\,dx$

Question 38.

a) Write $13-6x+x^2$ in the form $p+(x+q)^2$ where p and q are integers.

b) Hence, using the substitution $\frac{x-3}{2}=\tan u$, find $\displaystyle\int \frac{1}{13-6x+x^2}\,dx$.

Question 39. Find:

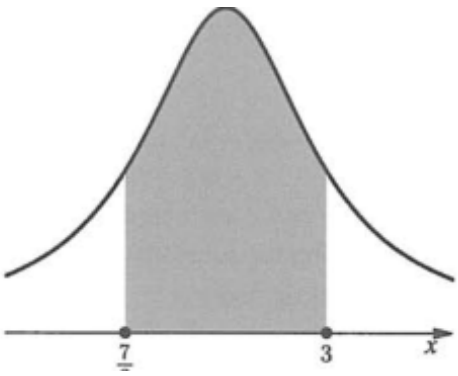
a) $\displaystyle\int \frac{1}{26+8x+2x^2}\,dx$

b) $\displaystyle\int \frac{6}{5+12x+8x^2}\,dx$

Question 40. The curve shown in the diagram below has equation

$$y = \frac{2}{65-48x+9x^2}.$$

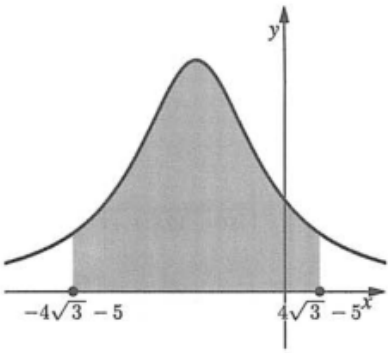
Find the exact area of the shaded region.



Question 41. The curve shown in the diagram below has equation

$$y = \frac{7}{41 + 10x + x^2}.$$

Find the exact area of the shaded region.



Question 42. Complete the square for each of these:

a) $-8 + 6x - x^2$
b) $-48 - 14x - x^2$
c) $-15 + 16x - 4x^2$
d) $-224 - 90x - 9x^2$
e) $-32 + 12x - x^2$
f) $-128 - 48x - 4x^2$

Question 43.

a) Write $-63 + 16x - x^2$ in the form $p - (x + q)^2$ where p and q are integers.

b) Hence, using the substitution $x - 8 = \sin u$, find $\displaystyle\int \frac{1}{\sqrt{-63 + 16x - x^2}}\, dx$.

Question 44. Find:

a) $\displaystyle\int \frac{1}{\sqrt{-3 - 4x - x^2}}\, dx$

b) $\displaystyle\int \frac{1}{\sqrt{-80 + 18x - x^2}}\, dx$

Question 45.

a) Write $-35 - 24x - 4x^2$ in the form $p - q(x + r)^2$ where p, q and r are integers.

b) Hence, using the substitution $2x + 6 = \sin u$, find $\displaystyle\int \frac{1}{\sqrt{-35 - 24x - 4x^2}}\, dx$.

Question 46. Find:

a) $\displaystyle\int \frac{1}{\sqrt{-63 - 32x - 4x^2}}\, dx$

b) $\displaystyle\int \frac{1}{\sqrt{-99 + 100x - 25x^2}}\, dx$

Question 47.

a) Write $-27 + 24x - 4x^2$ in the form $p - q(x + r)^2$ where p, q and r are integers.

b) Hence, using the substitution $\frac{2}{3}x - 2 = \sin u$, find $\displaystyle\int \frac{1}{\sqrt{-27 + 24x - 4x^2}}\, dx$.

Question 48. Find:

a) $\displaystyle\int \frac{1}{\sqrt{-11 - 36x - 9x^2}}\, dx$

b) $\displaystyle\int \frac{1}{\sqrt{3 + 10x - 25x^2}}\, dx$

Question 49. Show that

$$\int_2^{9/2} \frac{1}{\sqrt{21+4x-x^2}}\,dx = k\pi,$$

where k is a rational number to be found.

Question 50. A curve has equation

$$y = \frac{1}{\sqrt{400+x^2}}.$$

Find the exact volume formed when the region between the curve, the x-axis, the y-axis and the line $x = 20$ is rotated through 2π radians about the x-axis.

Question 51. Show that

$$\int_{2/5}^{\infty} \frac{1}{4+25x^2}\,dx = k\pi,$$

where k is a rational number to be found.